The last step is often the hardest

Two current LMIC field experiments and one migration classic · two peer-reviewed economics-journal articles. The following article pages are edited reading excerpts, not complete replacements for the originals.

LITERATURE PULSE

All three studies treat policy as a sequence rather than a switch. Colombian procurement training moves women-owned firms through registration and bidding, but contract wins concentrate among firms already able to use the opportunity. Tanzanian teacher training changes classroom practice and student learning, while extra software and peer diffusion add little. In the Philippines, information, search help, and passports raise search and job offers without raising migration. **Literature trend:** applied work is increasingly mapping the full behavioral and institutional funnel. Intermediate outcomes reveal where an intervention works; a stubborn final outcome reveals the next binding constraint. The practical lesson is to design complements around the weakest transition, not simply add more intensity to an earlier stage.

01 CURRENT / WOMEN-OWNED FIRMS AND PROCUREMENT ACCESS · STARTS ON PAGE 2 · World Bank Policy Research Working Paper 11481 · October 2026 · randomized trial · open access

Breaking into Supply Chains: Experimental Evidence on Procurement Training for Women-Owned SMEs

Can training move women-owned firms from interest in corporate procurement to actual contracts? A phased randomized trial with 1,105 eligible Colombian firms finds gains in knowledge and readiness, a 10-point increase in portal registration, and an 8-point increase in bidding. Winning rises mainly among firms with prior procurement knowledge or experience, by 8–9 points; less-experienced firms sometimes revise their optimism downward. The funnel is steep—55% attend the masterclass, 36% reach bootcamps, and 11% mentoring—and differential attrition plus short follow-up limit conclusions about profits and jobs. The paper matters because it separates access to the procurement pipeline from the capabilities needed to convert access into contracts.

02 CURRENT / EDUCATION QUALITY AND TEACHER PRACTICE · STARTS ON PAGE 19 · Journal of Development Economics 181 · 2026 · peer reviewed · CC BY 4.0

Participatory Teaching Improves Learning Outcomes: Evidence from a Field Experiment in Tanzania

Can a short shift away from rote instruction improve learning in crowded, resource-constrained schools? The authors randomize 220 Tanzanian primary schools, covering 440 teachers and about 25,000 students, and use official national exams 18 months later. A five-day participatory-pedagogy program raises math scores by about 0.13–0.15 standard deviations and the share earning top grades by roughly 5 points. Adding laptops and content software produces no further student gains, and intended peer-teacher spillovers are near zero. Attrition is modest and robustness checks are broad, but selected districts and schools temper external validity. The contrast between measured effects and uniformly enthusiastic participant reports is a useful warning against evaluating programs by satisfaction alone.

03 CLASSIC / MIGRATION, JOB SEARCH, AND POLICY BOTTLENECKS · STARTS ON PAGE 34 · Economic Development and Cultural Change 64(2) · 2016 · peer reviewed classic · open World Bank edition

Unilateral Facilitation Does Not Raise International Labor Migration from the Philippines

Why do enormous international wage gaps produce so little migration? In rural Sorsogon, the authors randomize information, online job-search help, and passport assistance across a large sample. The full package raises overseas search by 16 points and roughly doubles job offers, yet has essentially zero effect on migration over two years. Random assignment and pre-specified controls make the null unusually credible, though migration is rare and the study cannot identify one dominant final barrier. The classic lesson is that easing several source-country frictions can move every intermediate outcome without moving the final one when destination-side demand, financing, family costs, and worker preferences remain binding.

Editorial note: references and appendices were omitted where possible; no included article contributes more than 15 text pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.

Policy Research Working Paper 11481

Abstract

Women-owned-or-led small and medium-sized enterprises (WSMEs) are underrepresented in public and corporate supply chains worldwide. There is growing interest in promoting SMEs' and WSMEs' integration into value chains, but limited causal evidence whether interventions to support their participation in procurement are effective. This paper conducts a phased randomized controlled trial to estimate the impacts of corporate procurement training for WSMEs in Colombia. The program increases procurement knowledge and readiness, resulting in a ten percentage point increase in registrations to corporate portals, and an eight percentage point increase in the likelihood of submitting a bid. Impacts on winning bids are concentrated in WSMEs with prior procurement knowledge or experience, whose probability of winning a contract increases by eight to nine percentage points. These impacts are approximately twenty percent of the control means. More broadly, the paper illustrates how complier-weighted treatment effects can be used to evaluate programs delivered through a funnel-shaped participation process, common in private sector development interventions.

This paper is a product of the Development Impact Department, International Finance Corporation. It is part of a larger effort by the World Bank to provide open access to its research and make a contribution to development policy discussions around the world. Policy Research Working Papers are also posted on the Web at http://www.worldbank.org/prwp. The authors may be contacted at rcassidy@worldbankgroup.org, dmckenzie@worldbank.org, kneelamraju@ifc.org . A verified reproducibility package for this paper is available at http://reproducibility.worldbank.org, click **here** for direct access.



The Policy Research Working Paper Series disseminates the findings of work in progress to encourage the exchange of ideas about development issues. An objective of the series is to get the findings out quickly, even if the presentations are less than fully polished. The papers carry the names of the authors and should be cited accordingly. The findings, interpretations, and conclusions expressed in this paper are entirely those of the authors. They do not necessarily represent the views of the International Bank for Reconstruction and Development/World Bank and its affiliated organizations, or those of the Executive Directors of the World Bank or the governments they represent.

Produced by the Research Support Team

Breaking into supply chains: experimental evidence on procurement training for women-owned SMEs*

Rachel Cassidy† David McKenzie ‡ Kalyan Neelamraju §

Authorized for distribution by Xavier Giné, Lead Economist, Development Research Group, Development Economics, World Bank Group

Keywords: Entrepreneurship, Firms, Gender, Field Experiments
JEL Classification Codes: C93, J16, L25, L26, O12

*The authors thank Raaga Akkineni and Katherin Martinez Gomez for excellent research assistance; Carolina Salazar Pardo, Anna Domingo Riu, Genith Fuentes Ortiz, Tatiana Vargas Ibañez, Anne Kabugi and Heather Kipnis for their dedication in managing the *Sourcing2Equal* program and outstanding collaboration with this research project; Ortenca Destani, Emelly Mutambatsere, Deepa Chakrapani and Pablo Fajnzylber for guiding and sponsoring this research on behalf of IFC's Development Impact Department; Amy Luinstra and Robin Mearns for guidance on behalf of the World Bank Group Gender Department; Ana Maria Torres for sponsoring on behalf of IFC's Regional Operations in Latin America and the Caribbean; Girum Abebe, Sanola Daley and Diego Ubfal for peer review; Universidad EAFIT (Escuela de Administración, Finanzas e Instituto Tecnológico) and CESA (El Colegio de Estudios Superiores de Administración) for their work as implementing partners; and Innovations for Poverty Action for excellent implementation of the midline and endline surveys. The *Sourcing2Equal* program was funded by the Women Entrepreneurs Finance Initiative (https://www.we-fi.org/) and the Government of Norway. IRB Approval was obtained from HML IRB Research and Ethics, project reference no. 2719. The analysis plan is registered on the AEA RCT registry at https://doi.org/10.1257/rct.14881-4.0

†International Finance Corporation (IFC) rcassidy@ifc.org
(Corresponding author) 2121 Pennsylvania Avenue NW, Washington, D.C., 20433, USA
‡World Bank dmckenzie@worldbank.org
§International Finance Corporation (IFC) kneelamraju@ifc.org

1 Introduction

Women-owned-or-led small and medium size enterprises (WSMEs) are underrepresented in public and corporate supply chains worldwide, earning less than one percent of procurement spending by large corporations and governments (Vazquez and Frankel, 2017). Public procurement alone represents 13% of GDP in low- and middle-income countries, and similarly 14% in high-income countries.[1] Winning a public procurement contract (Ferraz et al., 2015; Mensah et al., 2026), becoming a supplier to a multinational (Alfaro-Ureña et al., 2022), and broader customer acquisition (Ignaszak and Sedláček, 2026) have been shown to have positive impacts on firm growth and job creation. A broader lack of demand, and being situated in more crowded markets, has been found to be a binding constraint on the growth of many female-led firms (Hardy and Kagy, 2020). Hence, access to large, relatively stable demand via government and corporate supply chains may help catalyze WSMEs' growth.

WSMEs may face both demand- and supply-side constraints in being able to receive procurement contracts. Demand may be limited if buyers have preferences and procedures that favor existing and large suppliers, or if some buyers perceive WSMEs to have lower capacity. Nevertheless, many buyers may perceive commercial benefits from diversifying and increasing competition in their supply chains, such as greater resilience to supply chain shocks, lower costs, or increased product varieties. There may also be positive reputational effects among socially-oriented customers and investors. However, even when buyers seek to source from WSMEs, they may lack sex-disaggregated data, or struggle to find eligible WSMEs among their bidders. On the supply side, WSMEs may concentrate in sectors where there are fewer opportunities, and tend to be smaller and less formalized than male-owned SMEs. WSMEs may also lack specific information, networks, skilled personnel and access to capital needed to apply for and fulfill corporate procurement opportunities. Further,

[1] https://blogs.worldbank.org/en/developmenttalk/how-large-public-procurement. The World Bank Group (WBG) itself achieved a commitment to double its procurement spending with WSMEs by 2023, and has increased its commitment to 12 percent of procurement spending by 2030.

WSMEs may not submit bids if they believe they have low chances of success. There is a lack of rigorous quantitative evidence on interventions that can overcome these constraints, and increase WSMEs' participation in procurement.

This paper conducts a randomized controlled trial (RCT) to evaluate a procurement-readiness training for WSMEs in three major cities in Colombia, designed to alleviate these supply-side barriers. The training was run as part of the International Finance Corporation's (IFC's) *Sourcing2Equal* program.[2] The overall program works on the demand side with corporate buyers, to facilitate peer-learning and offer advice on how buyers can systemically integrate gender-inclusive sourcing strategies in their procurement operations. In parallel, the program works on the supply side to build the capacity of WSMEs. The capacity-building training uses a funnel approach, which is a popular model for entrepreneurial support organizations and policymakers to offer some services to many participants, and then extend more intensive and expensive services to progressively smaller subsets. The funnel consisted of three main, procurement-focused interventions: i) short online masterclasses, offered to a large sample of firms; ii) bootcamps, comprising an online and an intensive in-person component, available to firms that complete the masterclasses and make the time commitment; and iii) mentoring, for a smaller number of "star suppliers", that complete both the masterclasses and the bootcamps, and are selected by the implementing team based on potential.

This funnel approach presents several challenges to the standard RCT set-up. First, aiming to serve a large number of firms in the first phase, coupled with the limited number of eligible WSMEs (typically requiring five or more employees), meant that there were insufficient firms for an oversubscription design and pure control group. Instead, the team worked to recruit 1,105 eligible WSMEs, then used the staggered roll-out of the program to conduct a phased-in RCT. Firms were randomly assigned to enter the funnel in three waves, each roughly six months apart. Impacts are then measured using a midline survey conducted approximately six months after the first wave was treated (with waves two and

[2] https://www.ifc.org/en/what-we-do/sector-expertise/gender/inclusive-supply-chains/sourcing2equal-program.

three serving as the control group), and an endline survey twelve months after the first wave and six months after the second wave were treated (with wave three serving as the control group). This design, of course, limits the ability to measure longer-term effects.

Second, incomplete take-up is a desired feature of the funnel approach, rather than an inconvenience to be minimized as in a standard RCT. The different stages of intervention self-select firms on effort, and reveal more information that the project team can use to ensure that scarce resources are spent on the firms they think are most likely to benefit. However, selective funneling reduces statistical power, and makes it harder to isolate which components of the bundled package had the most impact.

In addition to estimating the standard intent-to-treat (ITT) effect of the average effect of being offered an early starting place for the entire intervention funnel, we illustrate how progress can be made in unpacking the elements of the funnel through several additional estimates. First, we look at selection into the later stages, bootcamps and mentoring, and present correlational evidence of how outcomes vary with intervention participation. Second, and most standard, we calculate local average treatment effects (LATEs). The assumptions needed for these estimates are plausible for dealing with a lack of participation in the online masterclasses, but they become increasingly less credible for measuring impacts for the groups making it through to the bootcamps and mentoring. Third, and more novel, we illustrate the use of complier-weighted average treatment effects (Joffe and Brensinger, 2003; Coussens and Spiess, 2021), which put more weight on individuals that are predicted to be more likely to reach later stages.

The standard ITT results show impacts of the training on all but one of the pre-registered primary outcomes. These primary outcomes were designed to capture the outcomes that the project both targeted and thought it would be feasible to see change in after six and twelve months. Specifically, after six months the training produced impacts on an index of procurement knowledge (effect size 0.24 over a control mean of 3.65), an index of procurement readiness (effect size 0.28 over a control meant of 3.24), the proportion of WSMEs registering

with corporate portals (effect size 10 percentage points over a control mean of 53 percent) and submitting corporate bids (effect size 8 percentage points over a control mean of 43 percent). These six-month impacts are not significantly different when measured again twelve months after the intervention, and most remain significantly different from zero. In pooled estimates to improve power, the training also produces significant impacts on winning bids (5 percentage points over a control mean of 30 percent). The only primary outcome with a negative but insignificant coefficient is an index of attitudes toward procurement.

When examining heterogeneity by pre-specified firm characteristics, the training produced positive, significant impacts on winning corporate bids among firms that had some baseline knowledge (effect size nine percentage points over a control mean of 30 percent), or had won at least one corporate bid at baseline (effect size eight percentage points). Firms with no baseline experience significantly *decreased* their index of attitudes toward procurement (effect size -0.14 index points over a control mean of 3.69). This decrease was anticipated by the project team, and is consistent with a realistic reassessment of fit following the training, at least at these firms' current stage of development.

There was a steep funnel: of the 1,105 eligible firms, 612 attended the masterclass (55%), 399 the bootcamps (36%), and 120 the mentoring (11%). LATE estimates of the impact of attending at least the masterclass hence give impacts approximately 1.3 times as large as the ITTs. However, in order to believe a LATE for the effect of attending the bootcamp, or mentoring, we would need to assume no effect of the previous phases. The complier-weighted treatment effects (CWTEs) do not require the same exclusion restrictions, but instead require us to be able to predict which firms are more likely to make it through the different stages of the funnel. In practice, while some baseline variables do have predictive power, it is difficult to predict precisely which firms will make it to the later stages — underlining the value of self-selection in the funnel. This finding highlights a partial tension for implementation and evaluation: complier-weighted treatment effects will be more precise when compliance can be easily predicted; but if compliance could be easily predicted ex ante, there would be less

need for a funneling process. Nevertheless, our CWTEs do suggest larger impacts for the types of firms more likely to comply fully and to get to the mentorship stage. For example, we estimate a CWTE of 17.2 p.p. on registering in a corporate portal, compared to an ITT of 10.3 p.p.; and a CWTE of 7.3 p.p. on the likelihood of winning a bid, compared to an ITT of 4.8 p.p.

This paper contributes to three main literatures. First, a large literature has documented how women's firms lag behind their male-owned counterparts in profitability and size, and explores different explanations for these gaps (Aterido et al., 2013; World Bank Group, 2019; Jayachandran, 2021; Ubfal, 2026). Much of this work has focused on informal and micro firms, with relatively less evidence on WSMEs. The firms in our sample are already much larger and more profitable than most female-owned enterprises, with a median of six employees and potentially higher capacity for job creation including for women. Yet, less is known about what can be done to help these larger firms grow.

Second, we contribute to a literature on the use of training, mentoring, and consulting for firms, as surveyed by McKenzie et al. (2025). A key insight for our study is that other work in Colombia has shown how group-based consulting can have lasting impacts on Colombian SMEs (Iacovone et al., 2022, 2026). *Sourcing2Equal*'s bootcamps and part of the mentoring build on the use of group-based approaches to working with SMEs. Another insight is that mentoring is likely to be more effective when directed at advanced firms looking to innovate and do something new, where their local networks may not have sufficient expertise. *Sourcing2Equal*'s mentors are selected and coached to provide specialized knowledge on the procurement process that women are likely to lack, and find harder to learn on their own.

Third and most directly, we contribute to a new literature on causal impacts of interventions specifically designed to support SMEs' or WSMEs' participation in procurement. The most related work is Hjort et al. (2020), who conduct an experiment in Liberia. Their study invites 1,192 SMEs (average size four employees) to a seven-day "sellership" training program, focused on contracting and selling goods and services to large corporations or

estimate a marginal cost for participants in the masterclasses and bootcamps of US$422, and in the mentoring of US$626. These figures imply that the marginal cost of an additional firm participating in the entire funnel of masterclass, bootcamp and mentoring was $1,048.

3 Sample, randomization and data

3.1 Sample

The research team worked with the project team to run an open call from June to September 2024, with the aim of registering as large a number as possible of interested and eligible WSMEs. WSMEs were recruited by the project team via channels including collaboration with chambers of commerce and NGOs, and social media campaigns via IFC's and EAFIT's accounts on LinkedIn and Meta platforms. Advertisements explained that the program would be free to participants. WSMEs were asked to complete a short online registration form, if they wanted to apply to be considered for the virtual masterclasses, in-person bootcamps in Bogotá, Medellín or Cali, and mentoring for star suppliers. The form also asked contact information, and the city in which the firm would prefer to attend in-person activities if invited. The open call was national, since the project team reasoned that some women entrepreneurs may be willing and able to travel to one of these three major cities for the in-person component, even if their firm was not based there. However, the collaboration especially with EAFIT based in Medellin, plus the natural geographic distribution of firms, meant that most WSMEs who registered were based in the three cities. The language did not promise WSMEs a place in any particular activity, and no dates were promised for any particular event.

The registration form also collected the information to determine if the firm would be classified as a WSME and hence eligible. We-Fi defines women-ownership as: either (i) being at least 51% owned by women or a woman; or (ii) meeting all the following three sub-criteria: (a) being at least 20% owned by a woman or women, (b) a woman is the most senior or

across cities. Approximately two weeks later, firms were re-contacted and informed of their eligibility and their training cohort assignment. Firms in wave one were sent invitations to training on specific dates, while firms in waves two and three were told of the date windows in which they would be invited, with calendar invitations being sent nearer the time.

To minimize the risk that firms in waves two or three would drop out from the intervention, or from responding to the subsequent midline and endline surveys, the project team ensured regular interactions with the firms. Monthly newsletters were sent out to all registered participants, unless they opted to unsubscribe. Newsletters included opportunities to participate in polls about the issues most important to firms, highlighting of other opportunities that were valuable but minimally related to the outcomes to be studied by the RCT (such as business card workshops and invitation to ESG events), and information on when the midline and endline surveys were being launched.

3.3 Data

In addition to the baseline registration form that women filled out online, the research team conducted midline and endline surveys. An independent survey firm administered the questionnaire, contacting respondents directly with an introductory text that stressed the firm's independence and the confidentiality of answers, to minimize response bias. All 1,105 initial eligible firms from the RCT sample were sought out at both midline and endline (even if they did not respond at midline). The research team also accessed de-identified monitoring and evaluation data collected by the project team, to measure participation.[7]

Figure 1 summarizes the timing of the midline and endline surveys visually. The midline survey was conducted in April 2025, after wave one had completed their masterclasses, bootcamps, and introductory mentoring sessions if applicable. Interviews were conducted over the phone, with a minimum of five attempts to reach each respondent, unless they

[7]Colombia's excellent national firm-level administrative data unfortunately cannot be used to study impacts on outcomes such as revenues here. Administrative outcomes are reported for each fiscal year, whereas this study's waves two and three are control groups and then treated all within the same fiscal year (2025).

answered the phone and indicated non-consent to participate.

The endline survey was conducted in October 2025, after wave two had completed the same activities. Interviews were again mainly conducted over the phone, with up to ten attempts to reach respondents, and follow-up on email in cases where phone numbers did not work or respondents did not pick up. In-person "mop-up" interviews were conducted in cases where respondents answered the phone or email and stated a preference to conduct the interview in person. In-person interviews were also conducted in a small number of cases where respondents did not answer the phone or email, or did not have a working telephone number or email address, but did have a physical address on file and consented when enumerators reached them in person.[8]

At both midline and endline, respondents were incentivized to participate in the survey by notices in the newsletters and consent script explaining that the data would help improve understanding of WSMEs' needs for future similar programs. Further, respondents were informed that all individuals completing the survey would be entered into a prize draw for one of five tablet computers per round, with the winners advertised in the following monthly newsletter (to assure respondents that the draw would indeed be conducted).

Despite these efforts, Table 2 shows that attrition was considerable. Altogether, 65% of the initial sample of 1,105 firms responded to the midline survey, and 57% to the endline survey. Similar studies in other settings report comparable attrition. For example, Hjort et al. (2020) have a similar baseline sample size of 1,192 firms, and resurvey 789 firms (66%) at first follow-up, and 628 (53%) at second follow-up. Similar or worse response rates are also seen in other established surveys of formal SMEs, as well as larger microenterprises and large firms. For example, Scur et al. (2024) document the World Management Survey achieving average response rates of 32% in countries where participation is not mandatory.

[8] In a small number of cases, a different senior female representative of the WMSME attended one activity compared to another — for example if the COO attended the masterclass but then the CEO attended the bootcamp. The survey firm was given a strict protocol of whom to contact within the company, prioritizing individuals who participated in the most advanced activities, to ensure they interviewed the individual most informed about the program and procurement activities.

plan were therefore procurement knowledge, attitudes and beliefs, registration in corporate supplier portals, submission and winning of corporate bids (see also Table 4).[9]

Procurement knowledge is measured as the sum of answers to four questions: i) a subjective question about self-assessed knowledge; then three questions with objectively right or wrong answers about ii) sources of procurement information, iii) types of procurement process, and iv) the definition of a value proposition.[10] Procurement attitudes (beliefs) are measured as the average of the responses to five items on a five-point Likert scale, where a higher score means a more positive attitude. The items are: i) I am confident that my firm can approach and engage with corporate buyers; ii) I am confident that my firm can successfully compete for and win corporate procurement contracts; iii) corporations are genuinely interested in contracting with WMSMEs for social reasons; iv) corporations are genuinely interested in contracting with WMSMEs for commercial reasons; and v) winning corporate contracts is about personal relations, and not about the quality of the proposal (reverse coded). Procurement readiness is measured as the average of the responses to six questions about the firm's activities in the last six months on a five-point Likert scale, where a higher score means more effort is made to be ready for procurement contracts. The items are: i) conducted market analysis or search focusing on corporate procurement opportunities; ii) improved understanding of the corporate procurement process; iii) discussed or interacted with new corporate buyers and explored potential business opportunities; iv) discussed or interacted with other women entrepreneurs to explore synergies or learnings regarding cor-

[9]Outcomes pre-registered as secondary included number of corporate bids submitted and won, revenues from corporate bids, and employees hired. It was anticipated that a longer timeline would be needed to show impacts, while WMSMEs bore any immediate fixed costs of engaging in more corporate procurement, and as corporate contracts started to accumulate. Further, these continuous outcomes are noisier in nature, and ex ante power calculations suggested only large effects would be detectable. It was also anticipated that the intervention was likely to produce positive spillovers into firms' ability to compete in other forms of procurement, but over a longer time-frame. Other pre-registered secondary outcomes were therefore submission and winning of bids to other MSMEs, and to public buyers (see also Table 5).

[10]A larger set of questions with objectively right or wrong answers was included at midline, but some appear to have been too complicated to answer over the phone (e.g., requiring multiple right answers without prompting). At endline, we therefore reduced the number of objective questions and simplified the way in which they were scored. We are able to apply the endline scoring near-perfectly to the midline data, so we create a preferred midline measure "knowledge 0-9" which matches the measure constructed for endline. We also construct "knowledge -1-12" as pre-specified only at midline, available on request.

ω_s are training city x firm size (SME vs eligible micro) x having some baseline procurement knowledge x having won at least one corporate bid at baseline.[11]

Table 4 reports the results for the primary outcomes. There are positive six-month impacts on the index of procurement knowledge (effect size 0.24 over a control mean of 3.65, p-value 0.016), the index of procurement readiness (effect size 0.28 over a control meant of 3.24, p-value < 0.01), the proportion of firms registering to portals (effect size 10 percentage points over a control mean of 53 percent, p-value 0.01) and submitting corporate bids (effect size eight percentage points over a control mean of 43 percent, p-value 0.008).

These impacts are not significantly different after twelve months — and remain significant for procurement readiness, registration to corporate portals and submitting corporate bids. However, the twelve-month estimates are less precise, given the identifying variation comes from the wave one versus wave three comparison at endline only, and the impacts for procurement knowledge become insignificant. The point estimate for the impact on procurement attitudes is negative at both six and twelve months, but is not significant. Table 4 also presents tests that the six-month and 12-month effects are the same ($\beta_6 = \beta_{12}$). The tests fail to reject equality for all of the primary outcomes, although the test for equality of the six- and twelve-month impacts on procurement readiness has a p-value of 0.10.

Online Appendix Table A2 shows that the impact on knowledge is driven both by self-assessed knowledge, and the test question on knowledge of sources of procurement information — with a smaller, insignificant increase in the test question on knowledge of procurement

[11]The pre-analysis plan specified that we would seek further to improve power and reduce any imbalances due to attrition by using post-double LASSO to select additional covariates from the following baseline list, partialling out the randomization strata and the lagged outcome variable where available: number of employees (winsorized at the 99th percentile), log income in 2023, age of the company (winsorized at the 99th percentile), previously registered in a corporate database, number of previous corporate database registrations (winsorized at the 99th percentile), level of previous procurement knowledge, has an existing corporate client, number of corporate clients (winsorized at the 99th percentile), income from procurement contracts above median, sector (dummies created for the three main sectors), dummies if the SMEs are primarily women-owned or women-led. However, based on recent literature about the relatively small precision gains of using post-double LASSO to select controls, we instead include the full list of these controls rather than relying on LASSO to select (Cilliers et al., 2024). Since the standard errors increase compared to the estimations with no controls, rather than the stated objective of increasing power, we present the version without controls as our preferred specification. Results are robust to including the full set of controls, see Table 10.

corporate bids. Here again, impacts on winning corporate bids are significant only for those with higher baseline experience (eight percentage points over the same control mean of 30 percent, p-value 0.04). Similarly, impacts on submitting bids are only significant for those with corporate clients at baseline (eleven percentage points, over a control mean of 56 percent). In contrast, firms with no baseline experience show a significant *decrease* in attitudes toward procurement reported above (-0.14 index points, over a control mean of 3.69, p-value <0.01). This finding was anticipated by the project team, and is consistent with a learning mechanism whereby less experienced firms perhaps realistically update their priors downwards — at least at their current stage of development. The attitudes index is significantly correlated with participation in procurement via registering to portals and submitting bids at endline, although we cannot confirm the direction of causality.[13]

4.5 Which barriers were overcome and which remain?

The impacts on knowledge and readiness are consistent with descriptive evidence from open-ended questions, where treated firms were asked to report what, if anything, they had changed in their business after participating in *Sourcing2Equal*.[14] Six months after their participation in the program, 32.2% of respondents mentioned changes to their strategy and sales pitch, for example improving their business-to-business value proposition. Internal process improvements were noted by 15.5%, such as strengthening capacity to fulfil larger orders. Improvements in ability to meet documentary and compliance requirement were referenced

[13]At endline, the correlation between the attitudes index and having registered to a portal in the last six months is 0.11 (p-value < 0.01) in the whole sample, or 0.14 (p-value < 0.01) in just the treatment group). The correlation between the attitudes index and having submitted a bid in the last six months is 0.11 (p-value < 0.01) in the whole sample, or 0.11 (p-value < 0.01) in just the treatment group.

[14]We used AI to identify the most commonly cited themes and lists of keywords relating to each theme. The LLM used was Claude Opus 4.5. The specific prompt was "Create a list of keywords for responses to changes in the WSME business. Only include observations for respondents who participated in the project. Categorize those into themes." We then reviewed, adjusted and consolidated manually. We then used Stata to categorize responses by these themes, again with manual review and adjustment. Responses were often counted under multiple categories, as many respondents noted more than one key change or interrelated changes. Similar to our preferred specification for estimating treatment effects, we segment this descriptive analysis of responses by the length of time since exposure to the intervention — allowing us to distinguish changes six months (wave one at midline and wave two at endline, N=459) and twelve months (wave one at endline, N=268) after treatment.

4.6 Participation and complier-weighted treatment effects

Since the intervention as evaluated during the experimental timeframe is a bundle of first masterclasses, then bootcamps for many participants, then mentoring for a smaller number of selected participants, it is not possible to identify causally the separate impact of each activity. There was a steep funnel: of the 1,105 eligible firms, 612 attended the masterclass (55%), 399 the bootcamps (36%), and 120 the mentoring (11%). More broadly, we would expect impacts to be greater for firms who attend more of the program. We conduct several exercises to provide suggestive evidence as to which components may have played the most important roles.

First, Online Appendix Table A8 presents the mean baseline and endline outcomes by highest activity completion in the treatment groups: none, masterclass, bootcamp, or mentoring. Firms who went on to participate in no activity or just the masterclass appeared slightly worse at baseline than the control group in terms of mean outcomes. Those who went on to participate in the bootcamp (but not the mentoring) appeared approximately equal to the control group. Those who went on to participate in mentoring appeared much stronger at baseline than the control group, which was expected due to the program having selected mentoring participants based on potential observed after the bootcamp stage. At endline, treatment-group respondents who did not attend any activities appear weaker than the control group across all outcomes. Participants in only the masterclass appear slightly weaker than the control group, but this pattern may be attributable to the negative selection as observed from baseline data. Participants in up to the bootcamp appear slightly stronger than the control group, despite having appeared the same on observables at baseline. Participants in the mentoring appear much stronger than the control group across all outcomes, except attitudes. This pattern may in part reflect the positive selection designed by the program, and observed in the baseline data, or in part the more resource-intensive nature of the mentoring producing greater impact.

last two years, whereas at midline and endline it asked about the last six months to align with treatment windows. Comparing outcomes among wave three respondents suggests no adjustment of behavior from being a control group far from treatment (at midline) to being a control group about to start treatment (at endline) — except attitudes appear to be worse at endline just prior to treatment, which if anything implies that the negative treatment effect may be attenuated. That said, comparisons of midline and endline control means may also be influenced by seasonality in the procurement cycle. At midline, wave two are significantly more likely to have submitted and won a bid than wave three (excluding Cali for comparability), suggesting that if anything control groups increase their bidding behavior prior to treatment — again, attenuating our positive observed treatment effects.

5 Conclusion

This study contributes to a relatively sparse evidence base on interventions to support SMEs, especially women-owned-or-led SMEs, and their participation in procurement. The RCT demonstrates that the project produced positive impacts on the outcomes it targeted: knowledge, readiness, registering to portals and submission of bids. Firms with baseline experience also became more likely to win bids. The fact that the vast majority of corporate bids won are with buyers who did not participate in the program suggests that the training builds suppliers' capabilities that open doors beyond the program's own buyer network. In contrast, firms with no baseline experience revised their attitudes toward procurement downward.

The phased-in design meant little chance of seeing impacts on longer-term outcomes such as revenues, profits, and jobs. It is also possible that impacts for firms with less baseline experience might take longer to materialize. A future study, in a context with a thicker market of WSMEs in sectors amenable to procurement, might permit registration of sufficient firms to generate oversubscription. Such a setting could then allow for a pure control group, to measure impacts over a longer time horizon. Future work could also test the

Table 4: Primary Outcomes — 6-vs-12-month

	6 mths		12 mths			
Outcome	Control Mean (1)	T Coeff. (2)	Control Mean (3)	T Coeff. (4)	N (5)	6=12mth pval (6)
Procurement Knowledge (0-9)	3.65 (1.78)	0.24** (0.10)	4.55 (1.55)	0.17 (0.14)	1,342	0.63
Procurement Attitudes (1-5)	3.65 (0.56)	-0.05 (0.03)	3.55 (0.56)	-0.08 (0.06)	1,338	0.48
Procurement Readiness (1-5)	3.24 (0.89)	0.28*** (0.06)	3.30 (0.91)	0.15* (0.09)	1,333	0.10
Registered to corporate portal in last 6m$^+$	0.53 (0.50)	0.10*** (0.03)	0.52 (0.50)	0.10** (0.05)	1,335	0.98
Submit corporate bid in last 6m$^+$	0.43 (0.50)	0.08*** (0.03)	0.43 (0.50)	0.07* (0.04)	1,315	0.93
Won corporate bid in last 6m$^+$	0.30 (0.46)	0.04 (0.03)	0.29 (0.45)	0.06 (0.04)	1,315	0.73

Notes: Baseline was conducted in June-September 2024, midline in April 2025, endline in October 2025. After baseline, participants were randomly assigned to three waves, which were treated in phases. At midline, Wave 1 is treated for six months and Waves 2 and 3 are control. At endline, Wave 1 is treated for 12 months, wave 2 is treated for six months, and Wave 3 is control. Estimates presented in this table use a stacked panel of midline and endline data with survey round fixed effects and strata dummies. +: ANCOVA regression estimates for outcomes captured at baseline. **Knowledge:** index of self-assessed knowledge, knowledge on procurement sources, procurement process, and value proposition. **Attitude:** index of confidence to engage buyers, successfully bid, large corps interest in MSMEs for social and/or corporate reasons, proposal quality wins bid. **Readiness:** index of conduct market study, improve bids process, explore connections (new and with women entrepreneurs), improve proposals, integrate feedback. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9: Primary Outcomes — ANCOVA vs LATE vs CWTE (Pooled Sample)

	ANCOVA*	LATE			CWTE			
			Participated			Participated		
Outcome	T Coeff.	Masterclass	Bootcamp	Mentorship	Masterclass	Bootcamp	Mentorship	N
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Procurement Knowledge (0-9)	0.224**	0.275**	0.377**	1.483**	0.256***	0.262***	0.262**	1,342
	(0.095)	(0.115)	(0.155)	(0.610)	(0.097)	(0.101)	(0.125)	
Procurement Attitudes (1-5)	-0.055	-0.067	-0.091	-0.360	-0.056	-0.061*	-0.061	1,338
	(0.035)	(0.042)	(0.058)	(0.233)	(0.035)	(0.036)	(0.041)	
Procurement Readiness (1-5)	0.246***	0.301***	0.410***	1.614***	0.257***	0.270***	0.308***	1,333
	(0.056)	(0.067)	(0.090)	(0.366)	(0.056)	(0.058)	(0.075)	
Registered to corporate portal in last 6m^{+}	0.103***	0.126***	0.173***	0.677***	0.117***	0.129***	0.172***	1,335
	(0.029)	(0.036)	(0.048)	(0.194)	(0.030)	(0.031)	(0.037)	
Submit corporate bid in last 6m^{+}	0.075***	0.092***	0.126***	0.491***	0.083***	0.089***	0.118***	1,315
	(0.028)	(0.034)	(0.047)	(0.183)	(0.029)	(0.030)	(0.037)	
Won corporate bid in last 6m^{+}	0.048*	0.058*	0.080*	0.316*	0.051*	0.050*	0.073*	1,315
	(0.027)	(0.033)	(0.045)	(0.174)	(0.028)	(0.030)	(0.038)	

Notes: +: ANCOVA Regression estimates for outcomes captured at baseline. LATE estimates instrument participation by treatment assignment. CWTE estimates predict participation using baseline observables and baseline analogues of the outcome variables, then reweight observations according to their predicted probability of participation. Baseline was conducted in June-September 2024, midline in April 2025, endline in October 2025. After baseline, participants were randomly assigned to three waves, which were treated in phases. At midline, Wave 1 is treated and Waves 2 and 3 are control. At endline, Waves 1 and 2 are treated and Wave 3 is control. Estimates presented in this table use a stacked panel of midline and endline data with survey round fixed effects and strata dummies. **Knowledge:** index of self-assessed knowledge, knowledge on procurement sources, procurement process, and value proposition. **Attitude:** index of confidence to engage buyers, successfully bid, large corps interest in MSMEs for social and/or corporate reasons, proposal quality wins bid. **Readiness:** index of conduct market study, improve bids process, explore connections (new and with women entrepreneurs), improve proposals, integrate feedback. Baseline obserables include numbers of employees, company age, income, corporate database registration, number of databases registered in, number of procured clients, income made form procurement, having some procurement knowledge, women led vs women owned, and sector. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Participatory Teaching Improves Learning Outcomes: Evidence from a Field Experiment in Tanzania*

Martina Jakob[a,°], Konstantin Büchel[a,°],
Daniel Steffen[b], Aymo Brunetti[a]

[a]University of Bern
[b]Lucerne University of Applied Sciences and Arts

September 5, 2023

Abstract

Participatory teaching methods have been shown to be more successful than traditional rote learning in high-income countries. It is, however, less clear if they can help address the learning crisis in low- and middle-income countries, where classes tend to be large and teachers have fewer resources at their disposal. Based on a field experiment with 440 teachers from 220 schools in Tanzania, we use official standardized student examinations to assess the impact of a pedagogy-centered intervention. A five-day in-service teacher training on participatory and practice-based methods improved students' test scores 18 months later by 0.15σ. The additional provision of laptops with a learning software allowing teachers to refresh their content knowledge did not yield further learning gains for students. Complementary results from qualitative surveys and interviews suggest that the program was highly appreciated by different stakeholders, but that participants are unable to assess its impact along different dimensions, giving equally positive evaluations of its successful and its less successful elements.

JEL classification: C93, I21, J24, O15.
Keywords: productivity in education, participatory teaching, teacher content knowledge, computer-assisted learning, development economics.

*The authors acknowledge generous funding by the IMG Stiftung. This study received IRB approval from the Faculty of Business, Economics and Social Sciences at the University of Bern on November 4, 2019 (serial number: 122019). A randomized controlled trials registry entry is available at: `https://www.socialscienceregistry.org/trials/4959`.
Contact Details (°shared first authorship):
°Jakob: Univ. of Bern, Inst. of Sociology, Fabrikstr. 8, CH-3012 Bern, martina.jakob@unibe.ch
°Büchel: Univ. of Bern, Dept. of Economics, Schanzeneckstr. 1, CH-3001 Bern, konstantin.buechel@unibe.ch
Steffen: Lucerne University of Applied Sciences and Arts, Suurstoffi 1, CH-6343 Rotkreuz, dani.steffen@hslu.ch
Brunetti: Univ. of Bern, Dept. of Economics, Schanzeneckstr. 1, CH-3001 Bern, aymo.brunetti@unibe.ch

1 Introduction

Only 4 percent of students in low-income countries, compared to 95 percent in high-income countries, reach minimum literacy skills towards the end of primary school (World Bank, 2018, p. 8). To narrow the global learning gap, we need to rethink the strategies that teachers in developing countries use in the classroom. While schools in high-income countries have increasingly adopted participatory pedagogical approaches with a high degree of student engagement, more teacher-centered approaches such as lecturing and rote learning are still the norm in many low- and middle-income countries. Modern pedagogy takes a clear stance and considers student engagement a vital component of effective teaching, a view that is corroborated by vast evidence from high-income countries (e.g. Cornelius-White, 2007; Seidel and Shavelson, 2007; Harbour et al., 2015). However, it is not clear if this insight can be transferred to low- and middle-income countries, where teachers often have to manage very large classrooms and have few teaching aids at their disposal. Under such constraints, switching to more demanding teaching strategies could even prove detrimental (e.g., Berlinski and Busso, 2017). Moreover, in light of recent evidence on insufficient subject mastery among many teachers in disadvantaged regions (e.g., Sinha et al., 2016; Bold et al., 2017a; Brunetti et al., 2023), it remains an open question whether improving pedagogy alone is effective or if shortfalls in teachers' content knowledge need to be tackled simultaneously.

To address these questions, we conducted a randomized controlled trial (RCT) with 440 math teachers and more than 25,000 students from 220 schools in Tanzania. With an average of 51 students per teacher and a persistent shortage of classrooms and teaching aids, Tanzania faces resource constraints that are typical for many education systems in low-income countries (UNESCO, 2022). The intervention we study consisted of a five-day in-service program where teachers learned how to engage their students more actively in classes, bring their teaching closer to every-day live, and collaborate in teams to handle large classrooms and exchange on teaching techniques. After the initial five-day workshop, all teachers were invited to half-yearly refresher meetings to revise concepts and discuss implementation issues. Half of the teachers in the treatment group were randomly selected to further receive a laptop with a computer-assisted learning (CAL) software enabling them to refresh their content knowledge. The learning software consisted of short math videos and quizzes from "Khan Academy", and teachers participated in additional sessions to familiarize themselves with the program and discuss their progress. Both versions of the treatment were administered by Swiss NGO Helvetas that has implemented teacher training programs in Tanzania since 2000.

We scraped student-level data from standardized assessments published by National Examinations Council of Tanzania (NECTA) to estimate the impact of the program on students, and used data from our own assessments to study intermediate effects on teachers. Our design allows us to analyze direct effects on participating teachers and their students as well as spillover effects on peer teachers and students in treated schools. To better understand the mechanisms behind potential effects, we complemented the experimental data with classroom observations, surveys, and in-depth interviews.

Our analysis establishes four sets of findings: First, switching to participatory pedagogy successfully improved overall student tests scores two years later by 0.15σ (p-value=0.018), and the share of students with top grades increased by 6 percentage points from 16 to 22 percent (p-value=0.013). Point estimates for pass rates are positive too, but do not reach statistical significance (p-value=0.117). These effects are particularly remarkable considering that we used data from official national tests

that were not specifically tailored to the intervention. Our complementary data shows that treatment teachers did indeed apply a wide range of the participatory pedagogical strategies taught in the training, such as group work (observed in 87% of classroom visits), games (28%) or dialogue (26%), and expressed great enthusiasm for the program in in-depth interviews.

Second, students who were taught by teachers equipped with laptops and CAL software did not outperform students whose teachers only participated in the pedagogical intervention. Point estimates for the difference between the teacher in-service training with and without supplying the CAL software are small and statistically insignificant. While teachers receiving the laptop with CAL software markedly improved their understanding of concepts related to the subdomain of number sense and arithmetic by 0.22σ (p-value=0.058), the effect on an overall score of math proficiency is statistically insignificant (p-value=0.135). The average teacher achieved 78 percent correct answers at baseline, suggesting that many teachers were already sufficiently proficient in their subject before the intervention.[1] This is in line with results from our heterogeneity analysis showing that the CAL based refresher was significantly more effective for teachers with low content knowledge at baseline.

Third, we do not find evidence for spillovers on indirectly exposed teachers and students in treatment schools, even though the program was specifically designed to produce such externalities. Although trained teachers and their peers self-reported that they engaged in cascading activities such as model lessons and peer learning groups, estimates for spillover effects at both the student and the teacher level are close to zero and statistically insignificant (p-value=0.403).

Fourth, the data from our complementary analyses allows us to compare participants' views about impacts of the program with the actual causal estimates from the RCT. We observe that participants' survey and interview responses are not very informative about what aspects of the program did or did not work, as respondents gave equally positive evaluations for all of them. For example, while 74 percent of the trained teachers strongly agree with the statement that the program improved their pupils' math skills, so do 78 percent of their indirectly exposed colleagues, even though we do not find any indication for such spillovers in our experimental data.

Our study contributes to a growing body of literature on how to address the learning crisis in developing countries. A vast spectrum of approaches has been evaluated in recent decades (see, e.g., Kremer et al., 2013; Glewwe and Muralidharan, 2016; World Bank, 2018, for an overview), but one key factor has received surprisingly little attention: teachers. Closing the global learning gap will crucially depend on how teachers in low- and middle-income countries perform in the classroom. The pivotal role of teachers in developing countries has been appreciated by recent studies focusing on the role of teacher incentives and pay, including De Ree et al. (2018), Duflo et al. (2012), Mbiti et al. (2019b), and Muralidharan and Sundararaman (2011). Yet, the teacher performance not only depends on the economic incentives instructors face, but also on the repertoire of teaching strategies they have at their disposal.

A common strategy pursued by many development agencies is the promotion of a more student-centered pedagogy. Our study provides support for this approach, suggesting that attending five days of training in participatory pedagogy can be enough for teachers to restructure their classes and achieve higher learning gains for their students – even when their classes are large and few teaching aids are readily available. Promoting more engaging teaching strategies in low- and middle-income

[1] It is noteworthy that this substantially higher than the performance of teachers in El Salvador who averaged 47 percent on an almost identical assessment (Brunetti et al., 2020).

countries may thus be an essential element in the global quest for "inclusive and equitable quality education" (UN, 2015).

Our paper also ties into a nascent strand of literature studying complementarities in the educational production function (e.g., Mbiti et al., 2019a). Our findings suggest that shortfalls in teacher content knowledge are unlikely to constitute a binding constraint to effective teaching in Tanzanian primary schools. Teachers already exhibited considerable subject mastery, and the pedagogy intervention was at least equally successful in improving student learning without simultaneously addressing shortfalls in content knowledge.

We also add to the literature on treatment externalities. The canonical example for treatment externalities in education was documented by Miguel and Kremer (2004), where treating students with de-worming pills produced large spillovers on non-targeted children such as younger siblings. Such treatment externalities can drastically boost the cost-effectiveness of an educational program, a fact that has given rise to so called *cascading models* to deliberately include the promotion of spillovers in program designs. Our findings suggest that in the context of pedagogical interventions, achieving such externalities may not be straightforward. A possible explanation is that teachers need a considerable degree of (first-hand) exposure to the new teaching strategies to be able and willing to effectively restructure their classes.

Finally, this paper contributes on the methodological discussion on how best to evaluate programs (Banerjee and Duflo, 2009; Garbarino and Holland, 2009). While qualitative methods such as surveys and interviews provide important insights and fruitfully complement experimental data, our findings suggest that they may be ill-equipped to assess the impact of a program and distinguish between its successful and less successful elements. This highlights the importance of quantitative analysis to learn what actually works rather than relying on people's self-reports about it.

2 Context and Intervention

Our study is set in Tanzania, a lower-middle income country in East Africa. Tanzania's education system faces several challenges that are typical for developing countries. The massive expansion of schooling starting in the late nineties has put considerable strain on schools throughout the country, and resulted in shortages of teachers, classrooms and teaching materials. Consequently, the pupil-teacher ratio in primary schools stands at 51 students per instructor (UNESCO, 2022). In this context, the country has struggled to translate enrollment into learning. For example, about sixty percent of students in grade 3 are unable to read and understand a simple paragraph (Sumra et al., 2015). Learning outcomes crucially depend on what teachers do in the classroom. However, a recent study finds that only 36 percent of teachers in Tanzania possess the minimum pedagogical knowledge needed for effective teaching (Bold et al., 2017a).

The program we study in this paper was implemented by Helvetas, a large Swiss development organization focusing on building capacity in Africa, Asia, Latin America and Eastern Europe. Helvetas has been active in Tanzania for more than 50 years with projects in a broad range of fields including agriculture, youth employment, and education. After several years of piloting teacher professional development at small scale, Helvetas, the Tanzanian Teachers' Union (TTU), and the Ministry of Education jointly launched the SITT program (Inclusive School-Based In-Service Teachers Training) aiming at transforming pedagogy in Tanzanian classrooms. Prior to the experimental evaluation we

discuss in this paper, the program had already been rolled out in 1,430 schools throughout North-eastern Tanzania.

The aim of the program is to promote a more *student-centered approach* to teaching that fosters active participation among pupils. This involves activities such as group work or students taking turns with the teacher to explain concepts in front of the class. To make classes more accessible and relevant to students, teachers are encouraged to incorporate practical examples from everyday life. Through the use of inexpensive local materials such as berries, stones or toothpicks, teachers also learn how to address shortages in high-quality teaching aids. These strategies are conveyed to teachers and to the responsible government officials through a centrally organized five-day workshop. After the initial training, teachers are invited to participate in biannual two-day refresher meetings, where the application of the strategies is discussed and experiences are shared. As a guide throughout the school year, each teacher receives a comprehensive manual summarizing the teaching strategies.

In the spirit of a *cascading model*, participating teachers are also encouraged to share their knowledge with all other teachers in their schools through different collaborative activities. Most importantly, they are expected to invite their colleagues to model lessons to showcase the new teaching methods in action. Trained teachers also have to organize peer learning groups where their peers can discuss their impressions from the model lessons and share their experience with the new pedagogical techniques in their own teaching. Finally, teachers are encouraged to manage large classes as a team to promote cooperative behavior and joint learning. The implementation of the new teaching strategies and the cascading activities is overseen by government quality assurance officers and the Helvetas team through monitoring visits to targeted schools. As an indirect monitoring tool, teachers are added to a "WhatsApp" group where they are expected to share their experiences.

In 2020, the intervention was supplemented by additional activities to address potential shortfalls in teachers' content knowledge. In this context, half the teachers participated in an extended version of the program where they received a laptop equipped with a *computer-assisted learning* software. Learning materials included video content and short quizzes in Swahili produced by Khan Academy and were provided through the offline-first learning platform Kolibri developed by Learning Equality. Learning videos were typically around 5 to 10 minutes long and structured into three broad themes, *(i)* Number Sense and Elementary Arithmetics (NSEA, 80 videos), *(ii)* Geometry and Measurement (GEOM, 80 videos), and *(iii)* Data, Statistics and Probability (DSP, 11 videos). Videos were shared through a user-friendly interface and complemented with short quizzes. Each quiz drew on a basis of roughly 20 items that were presented in random order. Upon submitting an answer, users received instant feedback. The software tracked performance and awarded badges of success for quizzes with at least five correct answers. Previous studies have shown computer-assisted studying with Khan Academy to be effective at improving test scores of both students Büchel et al. (2022) and teachers (Brunetti et al., 2023).

3 Research Design

3.1 Sampling and Randomization

To assess the impact of the in-service teacher training, we conducted a randomized controlled trial with a sample of 220 public primary schools in the Tanzanian districts in of Mbulu DC, Mbulu TC,

Karatu and Siha, where the program had not been introduced yet. The implementing organization adopted a selection protocol similar to earlier implementation phases by excluding the best performing and the geographically least accessible schools in each district.

The experimental design allows to distinguish between *direct effects* on participating teachers and their pupils as well as *cascading effects* on peer teachers and their pupils. Specifically, selected schools nominated two teachers for the study: one *targeted teacher* for possible program participation and one *peer teacher* who was included for the estimation of spillovers. The selection of both targeted and peer teachers was done in coordination with the district education office and tied to the conditions *(i)* that both teachers should instruct math, and that *(ii)* the *targeted teacher* should teach math to sixth grade pupils in 2020 and seventh grade pupils in 2021. This procedure yielded a total sample of 440 teachers from 220 schools.

After the selection of schools and teachers, the research team randomly assigned each of the 220 schools to one out of three experimental conditions (see Figure 1):

- Pedagogy (65 schools, 130 teachers): Targeted teachers participated in the pedagogy training and were instructed to share their knowledge with their colleagues at their school.
- Pedagogy + Content (65 schools, 130 teachers): Targeted teachers participated in the pedagogy training and were instructed to share their knowledge with their colleagues at their school. They also obtained a laptop with computer-assisted learning software to self-study math.
- Control (90 schools, 180 teachers): Targeted teachers did not participate in any intervention activities.

Randomization was conducted after the nomination of teachers and the baseline data collection, and was stratified along three dimensions: district of school, baseline performance of pupils (i.e., school average in the standard 4 national examinations in 2018), and baseline performance of targeted teachers (i.e., math assessment conducted in November 2019).

3.2 Data

We rely on nationally standardized tests to measure effects on students, and conducted our own assessments to study intermediate effects on teachers. This experimental data is complemented with qualitative data we collected through classroom observations, surveys, and interviews in the treatment group.

Student assessments. The National Examinations Council of Tanzania (NECTA) conducts two standardized national student assessments that can be leveraged for this study: the *Primary School Leaving Examination* (PSLE) administered in grade 7, and the *Standard Four National Assessment* (SFNA) administered in grade 4. These yearly assessments are conducted with the entire student population in the respective grades and have high stakes: failing SFNA requires pupils to repeat grades, and passing PSLE is mandatory for admission in secondary school. Both assessments cover various subjects, but we rely on math scores for the main analysis. The math module in PSLE consists of 45 items that need to be completed in two hours, and SFNA includes 25 math questions students

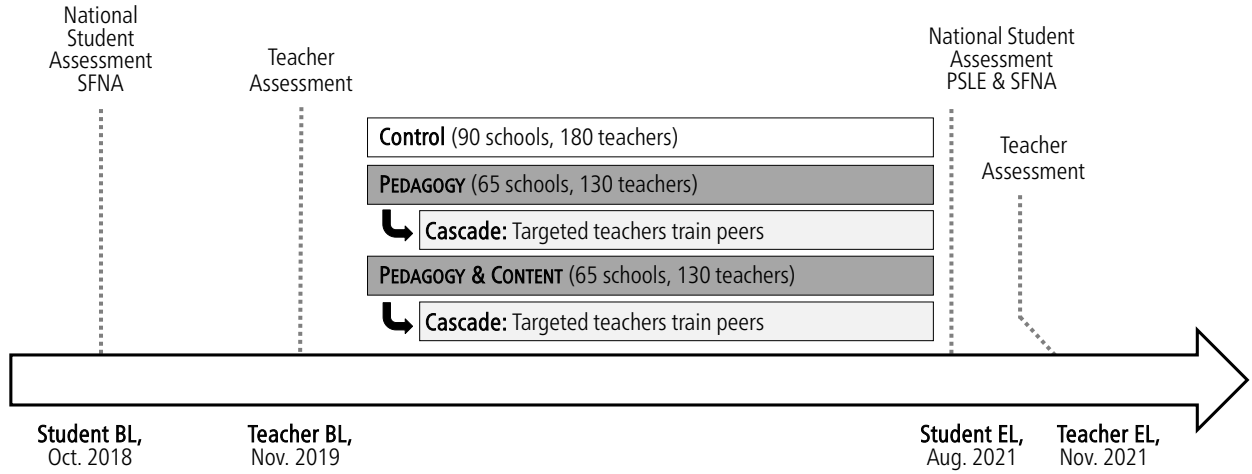


Figure 1: Timeline of the study.
The main intervention event is a five day workshop for all treated teachers and was conducted in February 2020. Afterwards, teachers implemented the new strategies and share them with their colleagues, participated in biannual meetings, and were visited by quality assurance officers of the Ministry of Education. The *National Standard Four Assessment* (SFNA 2018, SFNA 2021) and the *Primary Standard Leaving Examination* (PSLE 2021) are conducted by the Tanzanian government and the results are published online, see `https://onlinesys.necta.go.tz/`.
Source: Own representation.

need to answer in 90 minutes (NECTA, 2018, 2020). Assessment data is publicly available at the student level.

Our *main outcome measure* is the PSLE math score of seventh graders in 2021, the cohort taught by targeted (and potentially trained) teachers in 2020 and 2021. Pupils' PSLE scores can be merged with their SFNA scores from three years earlier (i.e., 2018) to establish a pupil-level baseline score. To assess spillover effects through *cascading*, the SFNA math scores from grade four pupils in 2021 can be used, as these pupils were taught by peer teachers in the same school who were exposed to cascading activities.[2] No baseline data is available in this case. As both PSLE and SFNA results are published online, we use web scraping to obtain the student-level data. Our final sample consists of 10,101 seventh graders to assess the direct effects of the programs and 15,023 fourth graders to estimate spillovers.

Teacher assessments. To measure teacher content knowledge in math, all 440 study participants were invited to two comprehensive math assessments conducted before and after program implementation. The assessments were designed to mirror the Tanzanian primary school curriculum between grade 2 and grade 7 and covered the domains Number Sense & Elementary Arithmetics (NSEA, about 60%), Geometry & Measurement (GEOM, about 35%), and Data, Statistics, & Probability (DSP, about 5%). Assessments were administered as paper-and-pencil tests in regional meet-ups and had to be completed in 90 minutes.

[2]Note that standard 4 pupils were not necessarily taught by the one peer teacher who was chosen to participate in the teacher assessments. However, this is irrelevant for the study of pupils' learning outcomes as cascading activities are explicitly targeted at all teachers in a school and hence should impact learning across all grades and classrooms in a program school.

Complementary qualitative data. We collected three different types of qualitative data to get deeper insight into how switching to participatory pedagogy was viewed and put in practice by treated teachers. First, all teachers had to fill in a short survey about their evaluations of the program and their perceptions about how it had impacted them and their students. The survey primarily included single-choice questions, where respondents could rate certain elements or indicate whether they agreed or disagreed with a given statement, but also featured space for written feedback and suggestions. Survey forms were administered during the endline math assessment to all teachers and tailored to the different experimental groups.[3] Second, to better understand how teachers incorporated the new methods into their classes, quality assurance officers of the education ministry conducted classroom observations in lessons of program participants. Based on the TEACH tool proposed by the World Bank (2019), a monitoring questionnaire was designed and government officials were briefed on how to conduct the classroom observations. Overall, 112 visits to treated teachers were conducted. Third, to complement the surveys and interviews, six participants of the PEDAGOGY intervention (about 120 min. audio recordings), six teachers from the PEDAGOGY & CONTENT group (about 120 min. audio recordings), six peer teachers (about 70 min. audio recordings), and twelve government or TTU officials (about 150 min. audio recordings) participated in *semi-structured interviews.*[4]

3.3 Baseline characteristics, compliance, and attrition

Table A.1 in the appendix shows that *baseline characteristics* are well-balanced across the three experimental groups. The average teacher in our sample scored 78 percent correct answers on the math test we administered prior to the intervention. As the test was designed to cover the Tanzanian primary school curriculum, this suggests that, on average, teachers master three quarters of the materials they have to teach. About 4 in every 10 teachers in our sample are female and the average teacher is 38 years old. Panel 2 on school characteristics shows that the typical class size is about 40 students.[5] The number of students that took the SFNA exam, roughly 50 per school, provides a proxy for the number of students per grade. As this figure is not much higher than the average class size, most schools can be assumed to have only one class per grade. Most importantly, pupils' baseline scores are well-balanced across experimental groups. On average, about 67 percent of students passed the baseline math exam, and 40 percent of students scored one of the two top grades (A or B).

Our monitoring data suggests that *compliance* with the treatment assignment was very high. All

[3]We designed four different questionnaires: (1) a questionnaire for teachers in the PEDAGOGY treatment with items about the training and the implementation of the new methods, (2) a similar questionnaire for the PEDAGOGY + CONTENT group with additional questions about the content training with the laptops, (3) a questionnaire for peer teachers asking about casacading activities, and (4) a short questionnaire for the control group with questions about the evaluation process. With the exception of the control group, the different survey versions followed the same basic structure and had many common items, allowing for comparison across different groups.

[4]During these conversations, the interviewees were asked *(i)* to share their general impression of the intervention, *(ii)* to explain their view on the main elements of the PEDAGOGY intervention, *(iii)* to share their assessment on the impact of the program on teachers' math and teaching skills as well as the learning outcomes of children, and *(iv)* to give feedback on selected activities and program inputs; additionally, officials were asked *(v)* to compare the pedagogical intervention with similar educational initiatives by other organizations, and *(vi)* to comment on their attitude towards rigorous program evaluation. Table D.1 in the appendix section D provides an overview of statements by topic and type of interviewee.

[5]While information on the number of pupils per classroom is difficult to collect, the number of pupils per *stream* can serve as a proxy. In Tanzania the concept of a "class" is surprisingly blurry because several streams of pupils can be instructed in one classroom (and effectively become one class) if schools do not have enough classrooms or teachers to teach streams separately.

teachers in the treatment group participated in the five-day teacher training, and 94 percent of the teachers in the PEDAGOGY & CONTENT group report having used the laptops for content revision. To be able to assess the impact of the program using students' tests scores in grade 7, targeted teachers had to teach math to all sixth graders in their school in 2020 and to all seventh graders in 2021. Our data collected during the endline teacher survey shows that 85 percent of the students in the treatment group were indeed taught by targeted teachers. This share does not differ significantly between experimental groups.

Tables A.2 and A.3 examine patterns of *attrition* for teachers and students respectively. At the teacher level, 99 percent of the selected teachers took part in the baseline assessment, and attrition for the endline assessment was about 15 percent and evenly distributed across experimental groups. This yields a total sample size of 368 teachers. At the student level, we start with baseline data for 12,657 pupils from 220 schools. About 17 percent of these students either dropped out of school between grade 4 and grade 7, missed the endline examination, or could not be matched between the two examination rounds. Moreover, one school dropped out because the targeted teacher missed both the base- and endline data collection. Finally, an estimation sample with 10,101 seventh graders from 219 schools remains. Both for teachers and pupils, attrition was unrelated to the experimental assignment. For the estimation of spillovers, we can use a sample of 15,023 grade 4 students from 220 schools. Due to the unavailability of baseline data, we cannot study the attrition for this cohort of students.

4 Results

4.1 Did promoting participatory teaching strategies improve learning?

We estimate the *intent to treat* (ITT) effect on students of directly targeted teachers with the following benchmark equation

$$Y_{isk}^{PSLE} = \beta Treatment_s + X_i^{'}\gamma + V_s^{'}\lambda + \phi_k + \epsilon_{isk}, \tag{1}$$

where Y_{isk}^{PSLE} is the standardized math PSLE score of student i in school s and stratum k at endline, and *Treatment* is a binary indicator that takes the value of 1 if a school was assigned to the treatment group and is 0 otherwise. Student level controls, X_i, comprise sex, baseline math score, and average baseline score across all subjects taken from the SFNA baseline assessment. V_s represents a vector of school-level controls including the number of students who took the baseline assessment, the average PSLE score at baseline[6], the driving distance to the district headquarters and the class size, as well as the math score, sex, and age of the targeted teacher. ϕ_k stands for k strata fixed effects, and ϵ_{isk} represents the error term.

The results in Table 1 document that students in treated schools significantly outperformed the control group by 0.15σ (column 2). Pupils in program schools were also up to 6 percentage points more likely to achieve a top grade (i.e., A or B) than their peers in control schools (columns 3 and 4). This corresponds to an increase in top grades by 36 percent. Estimates in columns 5 and 6 further

[6]Note that this is not the average score of the cohort we study, but that of a previous cohort of seventh graders in the school.

Table 1: Overall program effect on the math score of pupils

	Standardized		Scored A or B		Passed	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	0.107^{+}	0.145*	0.046*	0.056*	0.023	0.036
	(0.062)	(0.061)	(0.023)	(0.022)	(0.023)	(0.023)
Pupil baseline math score	0.466**	0.327**	0.121**	0.082**	0.210**	0.155**
	(0.017)	(0.021)	(0.008)	(0.008)	(0.008)	(0.010)
Mean of dep. variable	-0.008	-0.008	0.155	0.155	0.592	0.592
Observations	10101	10101	10101	10101	10101	10101
Adjusted R^2	0.252	0.295	0.146	0.180	0.202	0.224
Controls	No	Yes	No	Yes	No	Yes
Stratum fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The *dependent variable* is pupils' standardized math score for columns (1) and (2), a binary variable indicating whether a pupil scored A or B (highest grades) in math for columns (3) and (4), and a binary variable indicating whether a pupil passed the math exam for columns (5) and (6). *Pupil baseline math score* is a pupil's score in the SFNA exam administered in grade 4. Controls include *(i) pupil-level controls* for average SFNA baseline score across all subjects and sex, *(ii) school-level controls* for average PSLE baseline score (all subjects), class size, and number of pupils in grade 4 and *(iii) teacher-level controls* for sex, age, and math performance at baseline. Huber-White robust standard errors, clustered at the school level, in parentheses. + p<0.10, * p<0.05, ** p<0.01.

suggest that the program induced a 2 to 4 percentage point increase in pass rates, but these effects are not statistically significant at conventional levels.

Table A.5 in the appendix examines effects on students' average score across all subjects rather than their math score. Results are very similar, with estimated effects of 0.12σ and an increase in top grades by 7 percentage points or 30 percent. This suggest that although the pedagogical training was tailored to math, teachers were able to transfer the methods to other subjects.

Overall, the observed impacts are comparable to effects documented in RCTs of similar programs (see Snilstveit et al., 2015; McEwan, 2015). Unlike most other studies, our analyses are based on standardized national assessments that are not tailored to the intervention under study, which strengthens their external validity.

Our causal estimates are consistent with insights from our complementary data sources. Classroom observations point to a widespread use of the participatory teaching strategies advertised through the training program. As Figure C.1 in the appendix shows, treated teachers frequently applied methods such as group work (87% of visits), games (28%), student presentations (28%), and dialogues (26%). Treatment teachers also used a wide range of teaching materials, including daily life objects (66% of visits), textbooks (46%), and flash cards (20%). The survey data further shows that 96 percent of treated teachers rate the participatory teaching model as excellent (75%) or good (21%). Similarly, 96 percent of targeted teachers strongly (74%) or rather agree (22%) with the statement that the intervention improved their students' math scores. The high appreciation for the program also surfaced in the interviews where teachers often used words such as *"improve"*, *"change"*, and *"enjoy"* when talking about the intervention (see Table D.1 in the appendix).

To better understand under which circumstances the participatory teaching methods promoted through the training work best, it is informative to take a look at how effects vary by characteristics

Table 2: Program effect on the math score of pupils by implementation version

	Standardized		Scored A or B		Passed	
	(1)	(2)	(3)	(4)	(5)	(6)
T1: Pedagogy	0.127	0.147*	0.056+	0.059*	0.024	0.033
	(0.081)	(0.071)	(0.029)	(0.026)	(0.028)	(0.026)
T2: Pedagogy & Content	0.086	0.142+	0.034	0.052+	0.022	0.039
	(0.072)	(0.073)	(0.026)	(0.027)	(0.029)	(0.028)
Pupil baseline math score	0.466**	0.327**	0.121**	0.082**	0.210**	0.155**
	(0.017)	(0.021)	(0.008)	(0.008)	(0.008)	(0.010)
T2 − T1	-0.041	-0.005	-0.022	-0.008	-0.002	0.006
	(0.090)	(0.075)	(0.033)	(0.028)	(0.033)	(0.030)
Mean of dep. variable	-0.008	-0.008	0.155	0.155	0.592	0.592
Observations	10101	10101	10101	10101	10101	10101
Adjusted R^2	0.252	0.295	0.147	0.180	0.201	0.224
Controls	No	Yes	No	Yes	No	Yes
Stratum fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The *dependent variable* is pupils' standardized math score for columns (1) and (2), a binary variable indicating whether a pupil scored A or B (highest grades) in math for columns (3) and (4), and a binary variable indicating whether a pupil passed the math exam for columns (5) and (6). *Pupil baseline math score* is a pupil's score in the SFNA exam administered in grade 4. Controls include *(i) pupil-level controls* for average SFNA baseline score across all subjects and sex, *(ii) school-level controls* for average PSLE baseline score (all subjects), class size, and number of pupils in grade 4, and *(iii) teacher-level controls* for sex, age, and math performance at baseline. Huber-White robust standard errors, clustered at the school level, in parentheses. + p<0.10, * p<0.05, ** p<0.01.

of classes, teachers and pupils. A key challenge for productive student engagement is posed by the typically very large classes in Tanzania. According to Table A.7, the impact of the interventions decreased with larger class sizes, but these effects are not statistically significant (columns 7 and 8). A further concern might be that the use of participatory teaching methods demands a high level of skills on the part of the teachers. We do not observe teachers' pedagogical skills, but their performance in the math test can serve as a proxy. Indeed, treatment effects appear to be larger for students who are taught by better-performing teachers (columns 5 and 6). Additional analyses by pupils' gender and initial performance levels do not point towards relevant effect heterogeneity along these dimensions.

4.2 Did the computer-based content training yield additional benefits?

We also estimate the effects of each program version separately, using

$$Y_{isk}^{PSLE} = \beta_1 T1_s + \beta_2 T2_s + X_i^{'}\gamma + V_s^{'}\lambda + \phi_k + \epsilon_{isk}, \tag{2}$$

where $T1_s$ is a binary indicator for the Pedagogy intervention, and $T2_s$ indicates whether a treated teacher's school was additionally assigned to the content training component, i.e. to Pedagogy & Content.

As Table 2 shows, we do not find that providing laptops for content revision in addition to the pedagogical training yielded further learning gains for students. If anything, the point estimate for the extended intervention is slightly lower, but this difference is not significant.

One possible interpretation is that teachers did not use or appreciate the laptops for the intended purpose. Our complementary data suggests otherwise. Teachers report spending an average of 5 to 6 hours per week with the learning software, and provide very positive evaluations of the computer-assisted learning component with 68 percent rating it as excellent and 20 percent as good. The same affirmative feedback surfaced in interviews, where teachers unanimously expressed strong appreciation for the laptops and reported using them frequently for content revision or to prepare their lessons.

Another possibility is that teachers did use the laptops, but failed to meaningfully improve their content knowledge with the software. Figure 2 and Table A.8 present estimates for the causal impact of each intervention on teachers' content knowledge in math. Although teachers in the laptop group markedly improved their understanding of concepts related to NSEA by 0.22σ (columns 5 and 6 in Table A.8), the effect on an overall score of math proficiency is smaller (0.15σ) and misses conventional levels of statistical significance (columns 1 and 2).

A plausible interpretation for these modest effects is that most teachers already possessed good mastery of the primary school curriculum to begin with. As indicated in Figure A.2, the average teacher was able to answer 78 percent of the questions on materials covered in grades 2 to 7 correctly. While targeted teachers scored an average of 81 percent, peer teachers scored only 74 percent, suggesting that schools selected particularly well-performing teachers for program participation. Overall, 50 percent of the teachers pass the threshold for subject proficiency – at least 80 percent correct answers – advocated by the World Bank (Bold et al., 2017a). Only 2 percent of all teachers answered less than 50 percent of the questions correctly. A comparison with results from an almost identical assessment conducted with teachers in El Salvador suggests that the Tanzanian teachers perform considerably better than their counterparts in El Salvador (see Brunetti et al., 2020).[7] Hence, it appears plausible that many Tanzanian teachers are already sufficiently proficient in math for effective teaching at the primary school level. In line with this argument, Table A.9 in the appendix points to considerable effect heterogeneity by teachers' initial ability level. Low-performing teachers markedly improved their content knowledge (0.51σ, p = 0.004, for teachers below the median) due to the intervention, but these effects decline significantly as teachers' baseline scores improve, and are close to zero for high-performing teachers (not shown).

Hence, from an impact evaluation perspective, the additional investment in the IT equipment for content revision clearly did not pay off. Although we provide suggestive evidence that low-performing teachers used the software to catch up with their better-prepared colleagues, we do not find that such gains were transferred to students.

4.3 Did the interventions produce externalities for indirectly exposed students and teachers?

To estimate spillovers on indirectly exposed fourth-graders rather than directly exposed seventh graders, we use the following slightly adapted version of equation (1)

$$Y_{isk}^{SFNA} = \beta Treatment_s + X_i^{'}\gamma + V_s^{'}\lambda + \phi_k + \epsilon_{isk}, \tag{3}$$

where Y_{isk}^{SFNA} is the standardized math SFNA score of student i in school s and stratum k at

[7] The average teacher in the El Salvador study scored 47 percent on a math test covering materials from grades 2–6 and only 14 percent of teachers achieved at least 80 percent correct answers.

Table 3: Cascading effect on the math score of pupils

	Standardized		Scored A or B		Passed	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	0.028	0.037	0.011	0.012	0.011	0.016
	(0.048)	(0.044)	(0.009)	(0.009)	(0.021)	(0.019)
School PSLE avg. score (std)	0.081*	0.083**	0.016**	0.018**	0.034**	0.033**
	(0.031)	(0.031)	(0.006)	(0.006)	(0.013)	(0.012)
School SFNA avg. score (std)	0.134**	0.129**	0.022**	0.022**	0.048**	0.046**
	(0.031)	(0.031)	(0.006)	(0.006)	(0.013)	(0.013)
Mean of dep. variable	-0.000	-0.000	0.075	0.075	0.368	0.368
Observations	15023	15023	15023	15023	15023	15023
Adjusted R^2	0.072	0.080	0.035	0.040	0.053	0.060
Controls	No	Yes	No	Yes	No	Yes
Stratum fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The *dependent variable* is pupils' standardized SFNA math score for columns (1) and (2), a binary variable indicating whether a pupil scored A or B (highest grades) in math for columns (3) and (4), and a binary variable indicating whether a pupil passed the math exam for columns (5) and (6). *School-level baseline scores* are the school's average scores in the SFNA exam administered in grade 4 and the PSLE exam administered in grade 7. Controls include *(i) pupil-level controls* for sex, *(ii) school-level controls* for the number of pupils in grade 4 and *(iii) teacher-level controls* for sex, age, and math performance at baseline. Huber-White robust standard errors, clustered at the school level, in parentheses. + p<0.10, * p<0.05, ** p<0.01.

Hence, a more likely explanation is that although the cascading activities were conducted, they did not provide sufficient exposure to the new pedagogical techniques for peer teachers to effectively restructure their classes.

4.4 How informative are participants' self-reports about the impact of different program aspects?

An ongoing debate in the development community concerns the merits of two distinct evaluation traditions: a quantitative paradigm emphasizing causal inference methods and a qualitative tradition focusing on the experiences of project stakeholders (e.g., Banerjee and Duflo, 2009; Garbarino and Holland, 2009). The main contribution of this paper is quantitative, but we can also combine and compare our experimental findings with insights from qualitative surveys and interviews with project beneficiaries. In particular, we asked all participating teachers to assess the effect of the intervention on different outcomes, allowing us to contrast these self-reports with the actual causal effects we identified through the experiment (see Table 4).

Across all the outcomes and groups we study, participants are very confident about the impact of the intervention. While this is in line with the positive causal impact we report, response patterns appear to be unrelated to the success and failure of different project components. Most notably, directly participating teachers and peer teachers are equally optimistic about the impact of the intervention on their math skills and those of their students, even though we find no indication for spillover effects in our data. Similarly, we report no effect of the Pedagogy intervention on teachers' math skills, but 87 percent of teachers in this group strongly agree with the claim that they improved these skills

Table 4: Comparison between observed causal effects and participants' reported beliefs

	RCT: Observed impact	**Survey: Participants' beliefs about impact**
Impact of intervention on student learning	Significant effect of 0.15 SD*	Did the project improve the math skills of your pupils? Strongly agree: 74%, rather agree: 22%
Spillovers of intervention on students of peer teachers	Effect insignificant and close to zero	Did the project improve the math skills of your pupils? Strongly agree: 78%, rather agree: 19%
Impact of Pedagogy *intervention on teachers' math skills*	Effect insignificant and close to zero	Did the project improve your math skills? Strongly agree: 87%, rather agree: 5%
Impact of Pedagogy & Content *intervention on teachers' math skills*	Effect of 0.15 SD, but insignificant	Did the project improve your math skills? Strongly agree: 85%, rather agree: 11%
Spillovers of intervention on peer teachers' math skills	Effect insignificant and close to zero	Did the project improve your math skills? Strongly agree: 81%, rather agree: 15%

thanks to the intervention. Finally, teachers rated the self-studying with the laptops very positively, but we find only limited evidence for its effects at the teacher level and no evidence for an impact on students.

These findings tie into a nascent literature studying biases in evaluations (e.g., Camfield et al., 2014). Two broad explanations accounting for participants' overoptimistic impact assessments can be distinguished. First, people's capacity for counterfactual thinking is limited, leading them to misattribute outcomes or changes in their lives to the programs they participated in (e.g., McKenzie, 2018). Comparing actual and self-reported effects in three labor market interventions, Smith et al. (2021) conclude that participants act as "lay scientists". Their assessments are largely unrelated to the actual causal impact estimated for their group, but tend to follow coarse heuristics for this impact such as unconditional outcomes or before-after comparisons. A second well-documented bias in social science research, known as courtesy bias, social desirability bias or experimenter demand effects, is a general tendency of subjects to provide answers they perceive as aligning with the researcher's expectations (Camfield et al., 2014; Krumpal, 2013; Zizzo, 2010). In project evaluation, the resulting pro-project bias is likely to be exacerbated if people believe that the evaluation will determine whether the project is continued. Our findings are in line with these biases and suggest that while qualitative evidence from participant surveys and interviews can provide a valuable complement to experimental evidence, it is ill-equipped for the assessment of causal impacts.

5 Conclusion

Addressing the learning global learning crisis calls for innovative strategies to track and improve education (e.g. Patrinos and Angrist, 2018; World Bank, 2018; Jakob and Heinrich, 2023). In this paper we turn our attention to the teachers, who are the key actors in the educational system. While previous research has strongly focused on the misaligned economic incentives teachers often face, this study is premised on the assumption that they could be using ineffective pedagogy. Through a randomized controlled trial with 440 teachers and about 25,000 students in Tanzania, we show that promoting participatory teaching strategies significantly improves students' learning outcomes by 0.15σ. Our findings are based on standardized national assessments conducted by the National Examinations Council of Tanzania and corroborated by evidence from our classroom observations and participant surveys affirming that teachers indeed implemented and appreciated the new participatory methods.

Our study also explores the potential of computer-assisted learning to improve teachers' content knowledge and, thereby, student learning. We find suggestive evidence that providing computers with a learning software helps low-performing teachers improve their math skills. However, this does not translate into measurable learning gains for their students. Previous research suggests that a 0.1σ gain student learning would require a 1σ improvement in teachers' content knowledge (Bau and Das, 2020; Metzler and Woessmann, 2012) – an unrealistically large effect for educational interventions. Our findings underscore that addressing shortfalls in teachers' content knowledge is not a low-hanging fruit for promoting student learning.

We report similarly discouraging results for spillovers on other teachers and their students through cascading activities. Cascading schemes are favored in the development community for their potential to increase the number of beneficiaries and extend a project's reach. However, our results suggest that producing measurable learning spillovers is not straightforward. More research is thus needed to explore if and how the promise of cascading can be realized in educational initiatives.

Nevertheless, even without relying on spillovers, building teacher competencies can be a very cost-effective approach to improve student learning in the long run. Teachers often remain in their profession for many years, influencing dozens of student generations. If they continue to apply the new teaching methods throughout their professional lives, pedagogical teacher training becomes a highly sustainable and cost-effective means to foster student learning. Hence, promoting participatory teaching could be a key ingredient to a comprehensive strategy to ensure that children in developing countries are not only going to school, but are actually learning.

Policy Research Working Paper 6689

Abstract

Significant income gains from migrating from poorer to richer countries have motivated *unilateral* (source-country) policies facilitating labor emigration. However, their effectiveness is unknown. The authors conducted a large-scale randomized experiment in the Philippines testing the impact of unilaterally facilitating international labor migration. The most intensive treatment doubled the rate of job offers but had no identifiable effect on international labor migration. Even the highest overseas job-search rate that was induced (22 percent) falls far short of the share initially expressing interest in migrating (34 percent). The paper concludes that unilateral migration facilitation will at most induce a trickle, not a flood, of additional emigration.

This paper is a product of the Finance and Private Sector Development Team, Development Research Group. It is part of a larger effort by the World Bank to provide open access to its research and make a contribution to development policy discussions around the world. Policy Research Working Papers are also posted on the Web at http://econ.worldbank.org.
The authors may be contacted at dmckenzie@worldbank.org.

The Policy Research Working Paper Series disseminates the findings of work in progress to encourage the exchange of ideas about development issues. An objective of the series is to get the findings out quickly, even if the presentations are less than fully polished. The papers carry the names of the authors and should be cited accordingly. The findings, interpretations, and conclusions expressed in this paper are entirely those of the authors. They do not necessarily represent the views of the International Bank for Reconstruction and Development/World Bank and its affiliated organizations, or those of the Executive Directors of the World Bank or the governments they represent.

Produced by the Research Support Team

Unilateral Facilitation Does Not Raise International Labor Migration from the Philippines*

Emily Beam, David McKenzie, and Dean Yang

Keywords: International migration, passport costs, barriers to migration, unilateral migration policy, imperfect information, job-matching, field experiment, Philippines
JEL Codes: O15, F22, O15, C93

1. Introduction

Wage rates of workers using the same skills and doing the same jobs differ by as much as ten to one depending on the country in which they work (Ashenfelter, 2012). Moving from a developing to a developed country results in immediate large increases in income for the migrants, with gains that far exceed those of any other development policy intervention (Clements, Montenegro, and Pritchett, 2008; Hanson, 2009; McKenzie, Gibson, and Stillman, 2010; Gibson and McKenzie, forthcoming). Why do so few people emigrate, and what policies can governments in developing countries pursue to make it easier for their citizens to escape poverty through international migration?

Migration-source country governments have pursued two broad approaches to facilitating international migration for formal, legal work. Source countries can pursue *unilateral* facilitation policies on their own, without needing the cooperation of governments of migration-destination countries. Unilateral facilitation may involve provision of information, loan facilitation, and policies to ease the international job-search process. These policies act on the supply side of the migrant labor market. Enhanced unilateral facilitation could have positive impacts on migration if immigration policies in destination countries are sufficiently open, or if bilateral policies are already in place. *Bilateral* facilitation policies, on the other hand, involve cooperation with governments or employers in destination countries, and include formalization of agreements to allow labor migration of specified numbers and types of workers. Such policies primarily attempt to influence the demand side of the migrant labor market, but could also have supply-side components.

The Philippines has made perhaps the greatest progress among migration-source countries in implementing bilateral approaches, as evidenced by the existence of 49 bilateral migration agreements with 25 destination countries (Center for Migrant Advocacy, 2012) and an annual deployment of more than 2.0 million overseas Filipino workers (OFWs) worldwide (CFO, 2012). Consequently, overseas remittances top US$25 billion annually, nearly 10% of GDP (BSP 2012). However, the Philippines is not alone in promoting international migration; countries such as Bangladesh, Sri Lanka, and India are looking to the Philippine government's efforts as a model for promoting and regulating international migration (Ray et al., 2007).[1]

A wider range of countries have attempted unilateral policies to ease the barriers preventing their citizens from migrating. For example, several Pacific Island governments such as Tuvalu have provided financing for seasonal workers wishing to migrate abroad (Bedford et al, 2010); a number of countries have made it easier for their citizens to obtain passports, one notable example being Nepal, which decentralized the process of passport issuance so that citizens no longer had to travel over mountain ranges to Kathmandu to obtain a passport (McKenzie, 2007); other countries, such as Armenia, have attempted to provide potential migrants with more information about the disadvantages of illegal migration and about possibilities for legal jobs abroad (IOM, 2009); and Egypt created a jobs website to try to better connect Egyptian jobseekers and employers abroad (Fandrich, 2009).

Despite the spread of these policies, there is currently little rigorous empirical evidence on the effectiveness of either unilateral or bilateral migration facilitation. We implement a randomized experiment measuring the impact of unilateral migration facilitation. Our

migration. Individuals were randomized into a control group that received no treatment, or into receiving one or more of the set of facilitation treatments.

2. Setting

The Philippines is a useful setting to study the impact of unilateral approaches. The Philippine government's extensive bilateral facilitation policies, along with strong international labor demand, have created many migration opportunities in the past few decades. The government directly encourages international emigration and regulates private labor recruiters. Numerous financial institutions provide financial to help potential migrants pay recruitment fees (O'Neil, 2004). In the Philippines, even with this infrastructure in place, and despite the fact that the country's per capita GDP (around US$2,000) is less than one-tenth of that in developed countries, most Filipinos do not migrate, and five in six families do not receive remittances from workers abroad.

While the Philippines stands out as a promoter of international migration, it is far from alone in doing so. The promise of remittances and their potential to spur economic development has similarly motivated developed and developing country governments to encourage workers overseas either directly, through bilateral arrangements, or indirectly, by providing favorable tax treatment and incentives to encourage remittances (Puri and Ritzema, 1999; World Bank, 2006). The type of temporary migration common in the Philippines – legal migration of an individual as a temporary worker – is common worldwide, with almost all OECD countries having temporary worker programs; it is also the dominant form of labor migration into the Gulf countries, and to Singapore, Malaysia and Japan.

We conducted our experiment in Sorsogon, a rural province 10-12 hours by bus from the capital, Manila, where most recruitment activities take place. Reflecting its relative poverty and isolation, the Bicol region (where Sorsogon is located) has relatively low participation in international migration. The region accounts for 5.8% of the Philippine population, but only 3.3% of the country's overseas worker deployments in 2011 (NSO, 2011).

3. Methods

Early in 2010, we randomly selected 42 barangays from six municipalities in Sorsogon Province in which to conduct the baseline survey.[3] We collected a household roster from each barangay that included a list of households, and we used these to set barangay-specific target sample sizes proportional to population. We targeted approximately 5% of the total population from each barangay, or roughly 26% of households. We sorted households randomly and selected the first listed households to be our target. When a household could not be located or had no eligible members, we replaced it with the next household on the list.

From each household, interviewers screened the first member they met who had never worked abroad and was aged 20-45. Subsequent to the baseline survey, we learned from recruitment agencies that most individuals over age 40 would not be eligible for overseas work, so we restrict our baseline sample to the 4,153 individuals ages 20-40 we interviewed.[4] Houses selected were typically far enough apart from each other that concerns about information spillovers are second order; to the extent that there were spillovers, our treatment estimates are lower bounds on the differential impact of more information. The passport assistance was only offered to the respondents themselves, and

so it is not subject to such spillovers. Appendix A.1 describes our project timeline and sampling procedure in greater detail.

Table 1 reports demographic characteristics of the sample from the baseline survey. 71% of respondents are female, reflecting the fact that women were more likely to be at home when our project staff visited the household., but also enabling us to target those most likely to benefit from a reduction in barriers to overseas migration. Unlike some other migrant-sending countries such as Mexico, India, and Bangladesh, where the majority of migrants are male,[5] migration from the Philippines is female-dominated; between 1992 and 2009, 60.6% of new hires for overseas work were women (McKenzie, Theoharides, and Yang, forthcoming). Respondents report relatively high educational attainment (69% have completed high school and 36% have completed at least some post-secondary schooling) but low levels of household income (averaging P7,800 pesos/month, or US$157) suggesting they may have high returns to working overseas.[6] 34% report that they are "interested" or "strongly interested" in working abroad.

We revisited respondents in 2012 to collect information on their overseas job-search knowledge, job-search behavior, and migration decisions. We ask whether and how respondents searched for work overseas between 2010-2012, and we classify respondents as having migrated if they obtained a job offer and migrated abroad during that period.[7] We successfully surveyed 90.8% of respondents or another member of their household at endline, and we find no evidence of differential attrition across treatment assignment (Table A2).[8] Our primary analytical sample consists of these 90.8% for whom we successfully fielded an endline survey of the respondent or a fellow household member. Among the 9.2% who could not be reached at endline in this manner, we fielded brief

"log" surveys of neighbors on international labor migration by the respondent, and inclusion of these log surveys raises our total endline response rate (for the "migrate abroad" outcome) to 98.5%. We show in Appendix A.5 that our estimated impacts on migration are robust to use of the full (98.5%) endline sample, which includes the log surveys.

3.1 Interventions

Information and website assistance

During the baseline survey, we randomly assigned respondents to a control group or to one of four treatment groups designed to improve their information about and access to overseas work opportunities (Figure 1). These groups were application information [T1], financial information [T2], application and financial information [T1] + [T2], and website assistance [T4]. The application information consisted of information on typical overseas costs; the steps needed to apply for work abroad; an advertisement to enroll in Pilijobs.org, an overseas job-finding website designed as part of this project;[9] and a list of ways to avoid illegal recruitment from the Philippine Overseas Employment Agency. Financial information consisted of typical placement fees for work abroad and a list of Manila-based financial companies that provide loans for placement fees.

To facilitate job-matching, we worked with several Manila-based overseas recruitment agencies and a Sorsogon microfinance NGO to develop a website, Pilijobs.org, to help respondents easily contact and apply with reputable recruitment agencies and to allow those agencies to directly post job opportunities that could be accessed by respondents. While several widely used job-finding websites for overseas work already exist in the Philippines, we developed a separate one to ensure that applicants would be put in

contact only with high-quality, properly licensed recruitment agencies and to track their enrollment and participation in the website. Five recruitment agencies used the site, both to post job listings and to review applicants, and we worked closely with them to obtain their feedback and to encourage their staff to use the website. Section A.2 includes additional details about Pilijobs.org

Website assistance [T4] was always assigned along with application and financial information ([T1] + [T2]). It consisted of a paper form respondents could use to enroll in Pilijobs.org, and interviewers provided help if requested. Interviewers returned to pick up completed forms, or respondents returned them to a nearby office. Project staff encoded and uploaded forms to the website.

Passport assistance

Based on feedback from our partner recruitment agencies during the first stage of the project, we determined that another potential barrier to overseas migration was difficulty accessing a passport. Agencies reported that because of difficulty and delays many individuals encounter when applying for passports, they prioritized applicants who already had passports. In mid-2011, we randomly assigned a subset of our sample to one of two treatments targeted to help respondents get passports for overseas work, which were cross-randomized with our initial treatments to generate 15 total treatment and control cells (Figure 1).

The first passport treatment, passport information [T3], provided respondents a flier on the importance of having a passport before applying for overseas work and the steps they could take to obtain a passport. The second passport treatment, passport assistance [T3]+[T5], involved the passport information treatment, plus a letter inviting respondents

We use the following specification to measure the impact of unilateral facilitation on job-search and migration:

$$Y_i = \alpha + \sum_{j=1}^{14} \beta_j D_i^j + B'\lambda + X'\delta + \varepsilon_i,$$

where Y_i is the outcome variable for respondent *i*, measured in the 2012 endline survey. D_i^j is a binary indicator equal to one if respondent *i* is assigned to combination *j* of application information [T1], financial information [T2], passport information [T3], website assistance [T4], or passport assistance [T5].

Vector *B* includes the barangay/client-status set of stratification cell fixed effects, along with an indicator for whether the respondent was randomly selected to be in the passport sample. The coefficient on this indicator would be non-zero if simply being interviewed in the passport sample affected our endline outcomes. (In practice, this coefficient is consistently close to zero and not statistically significant.) To increase the precision of our estimates, we also include a vector of pre-specified controls, *X*, for the following baseline characteristics: female (indicator); age (continuous); high school completion (indicator); some college or vocational training (indicator); college completion (indicator); interested in working abroad (indicator); willingness to take risks (0-10 scale); household income (in thousands of pesos); household savings (in thousands of pesos); whether the household has ever taken out a loan (indicator); asset ownership (normalized index of durable asset holdings); whether the respondent has extended family overseas (indicator); and whether the respondent has immediate family overseas (indicator). Missing covariate values are coded as zeros, and we include a set of missing value indicator variables.

H1: The massive gain in income possible from migration should result in high migration demand. Since the monetary gains from migration are likely to far exceed the monetary costs for most Filipinos (Clemens, Montenegro, and Pritchett, 2008), theory predicts most individuals will wish to migrate unless the disutility from moving is high. In fact only 33.9% of individuals say they interested or very interested in migration at baseline, and far fewer search for work overseas (5.1% of the control group) between survey rounds.

H2: Incomplete information prevents individuals from realizing the gains from migration. If individuals underestimate the gains from migration (McKenzie, Gibson, and Stillman, 2013) or overstate the costs, then some individuals for whom it is optimal to migrate will decide not to do so. Knowledge is clearly incomplete – at baseline, one-quarter of individuals responded with "don't know" to the typical wages and costs of work overseas for six common destination countries, and the responses given by those who do give an answer also suggest considerable inaccuracies. For example, half of those who did respond estimated they would earn the same wage or less in high-wage Canada as they would in low-wage Saudi Arabia. At endline, only 14.3% of the control group can name a lender who can finance migration costs and only 19.9% know where to go to apply for a passport. However, the information treatments alone do not result in higher rates of job search or international migration.

Figure 2 highlights means of key outcomes across a representative subset of treatments. We see the rate of overseas job search (5.3%) for the "All information" treatment is similar in magnitude, and not statistically different, from the 5.1% rate in the control group, and that only 1.1% of the "All information" group migrates abroad over the two year period. Table 2 provides regression estimates of the treatment effects for a broader

range of job-search and migration outcomes over the two-year period and confirms this lack of impact. Table 3 restricts the regression analysis to the subset of individuals who indicated that they were interested in migrating at baseline. In this subsample, information alone induces statistically significant increases (at the 10% level) in the likelihood of being invited to interview and attending an interview for work abroad, but there is no statistically significant impact of information alone on actual migration.

H3: Frictions in matching with recruiters limit international migration. Even if individuals have correct information and decide the gains from migration exceed the costs, they still need to match with a job abroad (Ortega, 2000; Lumpe and Weigert, 2009). The website treatment is intended to help individuals do this. Figure 2 shows that the combination of information and the website treatment ("All Information + Website") caused a substantial increase in the rate of search for work abroad, from 5.1% to 15.7%. The regression-adjusted estimate of this treatment effect from Table 2 is nearly identical, indicating a 10.8 percentage-point increase (statistically significant at the 1% level). Despite inducing substantially higher search effort, the treatment causes no additional migration abroad: the coefficient estimate in Table 2 column 8 is very small in magnitude and is not significantly different from zero. For the subgroup expressing interest in migrating at baseline, Table 3 shows the website and information combination resulted in a 20.1 percentage-point increase in job search and a 7.7 percentage-point increase in attending an interview (statistically significant at the 1% and 5% levels, respectively), but much smaller and statistically insignificant increases in the job offer rate (4.4 percentage points) and in the migration rate (2.3 percentage points).

H4: Documentation barriers prevent individuals from taking advantage of job openings abroad. Lack of a passport may prevent recruiters from even considering individuals for job openings or prevent some of those who receive job offers from taking up these offers. Our most intensive "Full assistance" treatment, which combines information, website assistance, and assistance obtaining a passport, results in a 21.7% job-search rate (Figure 2), but it is still far short of the 33.9% reporting interest in migration at baseline. Table 2 shows that this 16.0 percentage-point increase in job search over the control group rate is statistically significant at the 1% level, and it mainly reflects increased online search (column 2, increase significant at the 1% level), in addition to some additional search via other methods, such as attending job fairs (column 4, increase significant at the 5% level). The full assistance treatment also has positive impacts on job-interview invitations, interview attendance, and job offer receipt (columns 5-7, effects significant at the 10%, 5%, and 10% levels respectively), and these effects are large relative to control group rates (2.6%, 1.5%, and 1.7%, respectively). Despite these positive impacts on pre-migration outcomes, the treatment has no statistically significant impact on migration abroad: the point estimate is zero percentage points to the third decimal place (column 8).

Table 3 shows these effects are larger for the sub-group initially expressing interest in migration (for whom demand should not be the constraint), with a 26.7 percentage-point increase in job search, a 8.3 percentage-point increase in job-interview attendance, and a 7.4 percentage-point increase in the likelihood of receiving a job offer abroad (all statistically significant at the 5% level or less). However, there is still only a statistically insignificant 1.6 percentage-point increase in migration abroad. That is, our full package

Perhaps the most surprising result of our study is that, while our most intensive facilitation treatment delivers statistically significant increases in overseas job offers (that are large relative to control group rates), it has zero impact on actual overseas migration (over a two-year post-treatment window). This finding contrasts strongly with recent work on facilitating internal migration (Bryan, Chowdhury, and Mobarak, 2012), which has found small subsidies such as the cost of a bus ticket can have large impacts on internal job search and internal migration.

This lack of impact serves to further underline the point that demand for international migration on the part of developing-country residents is likely to be overstated – those induced by an intervention to receive actual job offers commonly reject those offers in the end. Our survey evidence on the reasons these jobs are declined fails to pinpoint a dominant reason behind such job-offer rejections. The most common reason, financial constraints (cited by nearly a quarter of job-offer decliners), does not distinguish whether individuals face actual financial constraints or whether they are indicative that the perceived benefits of migration do not exceed the perceived costs.

Together, these results indicate that unilateral facilitation policies related to information, job search, and documentation assistance are not sufficient to increase rates of international labor migration. We find evidence of multiple remaining barriers on both the supply side (relatively low interest on the part of potential migrants) and demand side (highly selective screening for interviews and job offers) for overseas work. Our findings indicate that policymakers aiming to expand access to migration, particularly for those in isolated areas, should not expect to achieve success if relying solely on unilateral migration facilitation, and brings to the fore the role of complementary bilateral

facilitation policies. Investigating the effectiveness of such bilateral policies is an important avenue for future research.

References

Ashenfelter, Orley. 2012. "Comparing Real Wages." NBER Working Paper Series, No. 18006.

Bangko Sentral ng Pilipinas. 2012. "Overseas Filipinos' Cash Remittances by Country, by Source." http://www.bsp.gov.ph/statistics/spei_pub/Table%2034.pdf

Bedford, Charlotte, Bedford Richard, and Elsie Ho. 2010. "Engaging with New Zealand's Recognized Seasonal Employer Work Policy: The Case of Tuvalu", *Asian and Pacific Migration Journal* 19(3): 421-45.

Bryan, Gharad, Shyamal Chowdhury, and A. Musfiq Mobarak. 2011. "Seasonal Migration and Risk Aversion." Working paper.

Center for Migrant Advocacy, 2012. "Bilateral labor agreements and social security agreements," http://centerformigrantadvocacy.files.wordpress.com/2012/06/bilateral-labor-agreements-and-social-security-agreements1.pdf.

Clemens, Michael, Claudio Montenegro, and Lant Pritchett. 2009. "The Place Premium: Wage Differences for Identical Workers across the US Border." HKS Faculty Research Working Paper Series, No. RWP09-004.

Commission on Filipinos Overseas. 2012. "2008-2012 Overseas Employment Statistics." http://www.poea.gov.ph/stats/2012_stats.pdf

Fandrich, Christine. 2009. "Facilitated International Egyptian Labor Migration and Development: The integrated migration information system as case study", The

Table 2: Impact of unilateral facilitation on overseas job search and migration.

	From 2010-2012, did the respondent search for work overseas by ...				*From 2010-2012, did the respondent ...*			
	Any way	Using Internet	Visiting recruitment agency	Some other way	Invited to interview	Attend interview	Receive job offer abroad	Migrate abroad
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
[T1] + [T2] +[T3] **"All Information"**	0.002	-0.008	0.001	0.013	0.027	0.025	0.004	0.004
	[0.025]	[0.013]	[0.019]	[0.015]	[0.023]	[0.021]	[0.016]	[0.011]
[T1] + [T2] + [T3] + [T4] **"All Information + Website"**	0.108***	0.103***	0.000	0.012	0.018	0.022	0.009	0.002
	[0.025]	[0.022]	[0.014]	[0.010]	[0.015]	[0.013]	[0.012]	[0.007]
[T3] + [T5] **"Only Passport Assistance"**	0.076**	0.007	0.031	0.054**	0.024	0.024	0.012	0.004
	[0.036]	[0.020]	[0.026]	[0.026]	[0.025]	[0.022]	[0.020]	[0.012]
[T1] + [T2] + [T3] + [T5] **"All Information + Passport"**	0.044	-0.004	0.038	0.027	0.043*	0.029	0.007	0.015
	[0.029]	[0.014]	[0.025]	[0.019]	[0.026]	[0.022]	[0.017]	[0.016]
[T1] + [T2] + [T3] + [T4] + [T5] **"Full Assistance"**	0.160***	0.147***	0.014	0.026**	0.030*	0.032**	0.027*	-0.000
	[0.028]	[0.024]	[0.016]	[0.013]	[0.017]	[0.015]	[0.015]	[0.007]
Sample Size	3,802	3,802	3,802	3,802	3,802	3,802	3,802	3,802
Control DV Mean	5.1%	1.0%	2.9%	1.4%	2.6%	1.5%	1.7%	0.9%
P-value, coefficients jointly zero	0.000***	0.000***	0.256	0.020**	0.557	0.549	0.922	0.691

**** p<0.01, ** p<0.05, * p<0.10*

Sample includes baseline respondents with completed endline survey. Stratification-cell status fixed effects and an indicator for not being in passport sample are also included. Covariates include age, gender, education, interest in work abroad, willingness to take risks, HH income and savings, whether HH has taken out a loan, whether any immediate or extended family member currently abroad, and an normalized durable asset index. Missing covariates are coded as zeroes with a binary flag included. Huber-White standard errors reported in brackets. Coefficients for all treatments reported in Table A7.